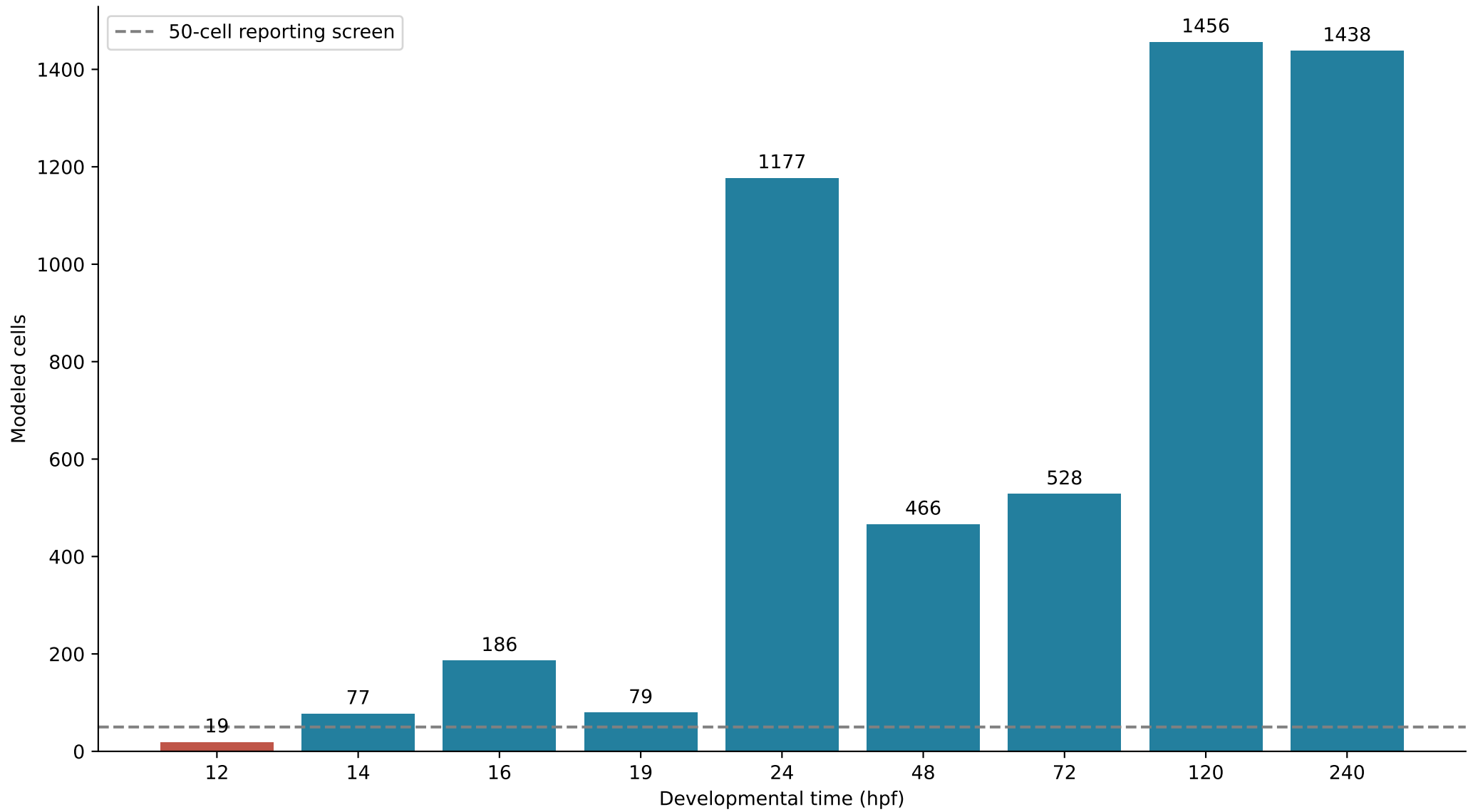
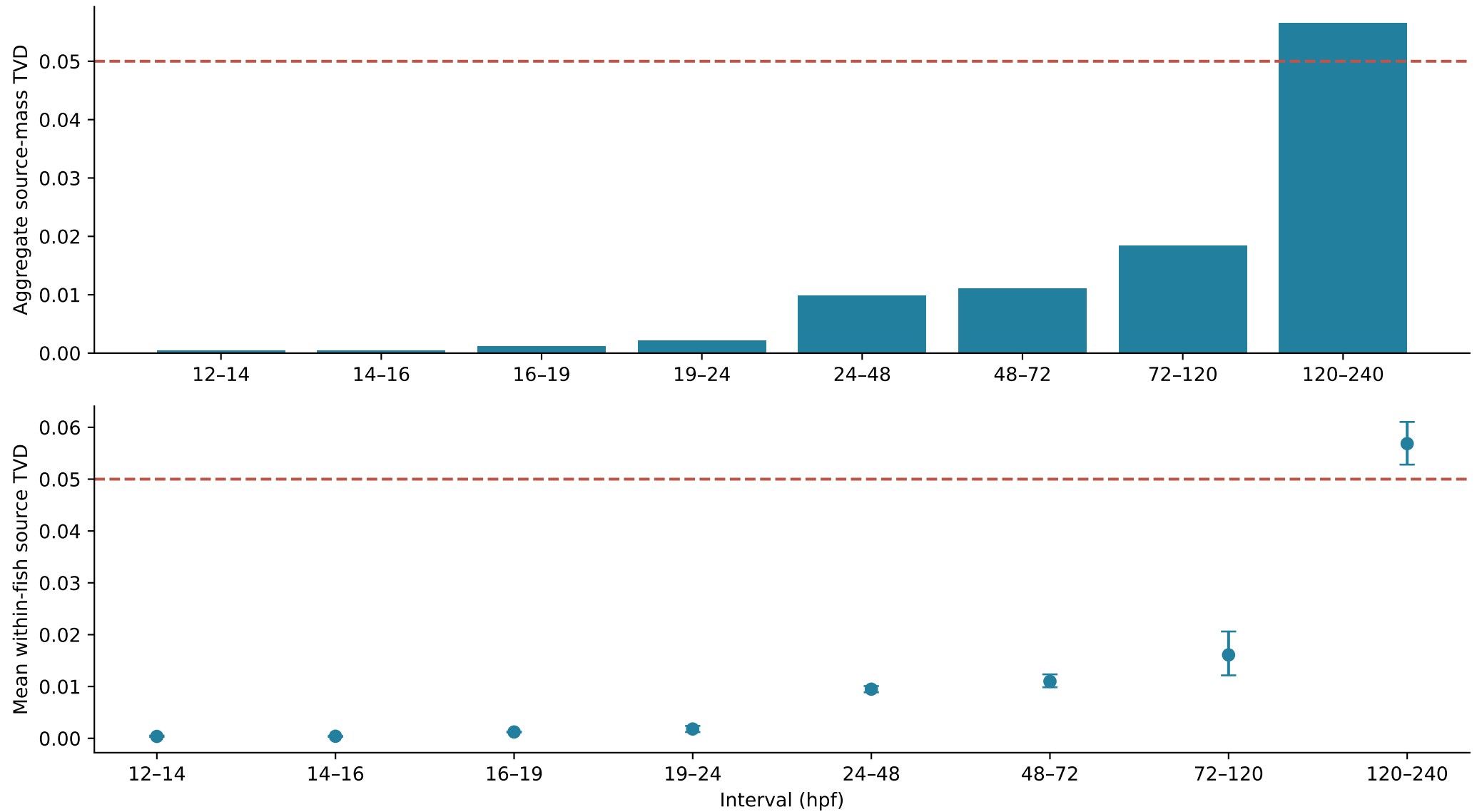


1. Cohort and sampling support



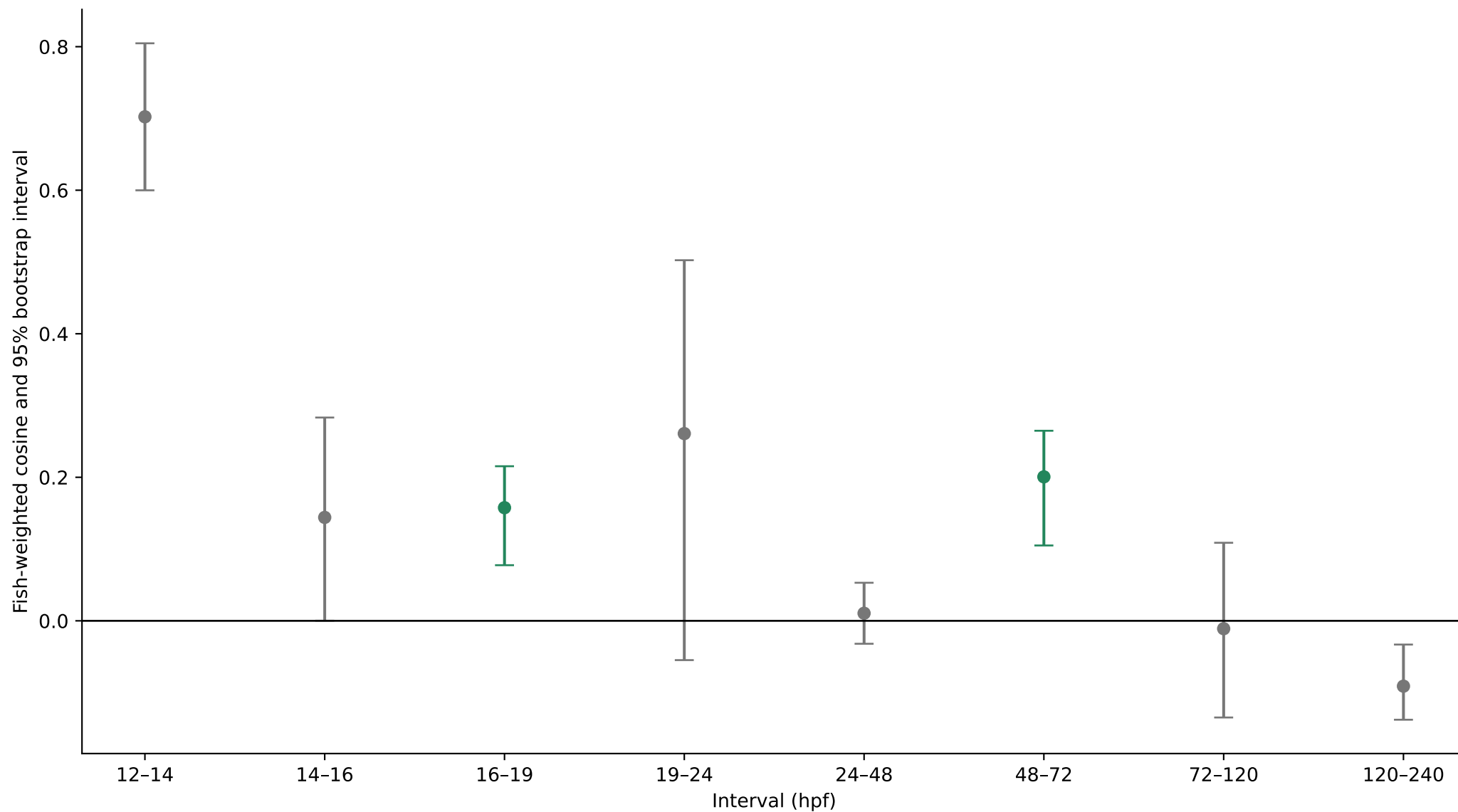
5,426 cells; 36 different fish; four fish per stage. The 12 hpf stage has 19 cells and cannot support type-level 12-14 hpf claims. Fish at different stages are different animals.

2. Growth priors change late transport most



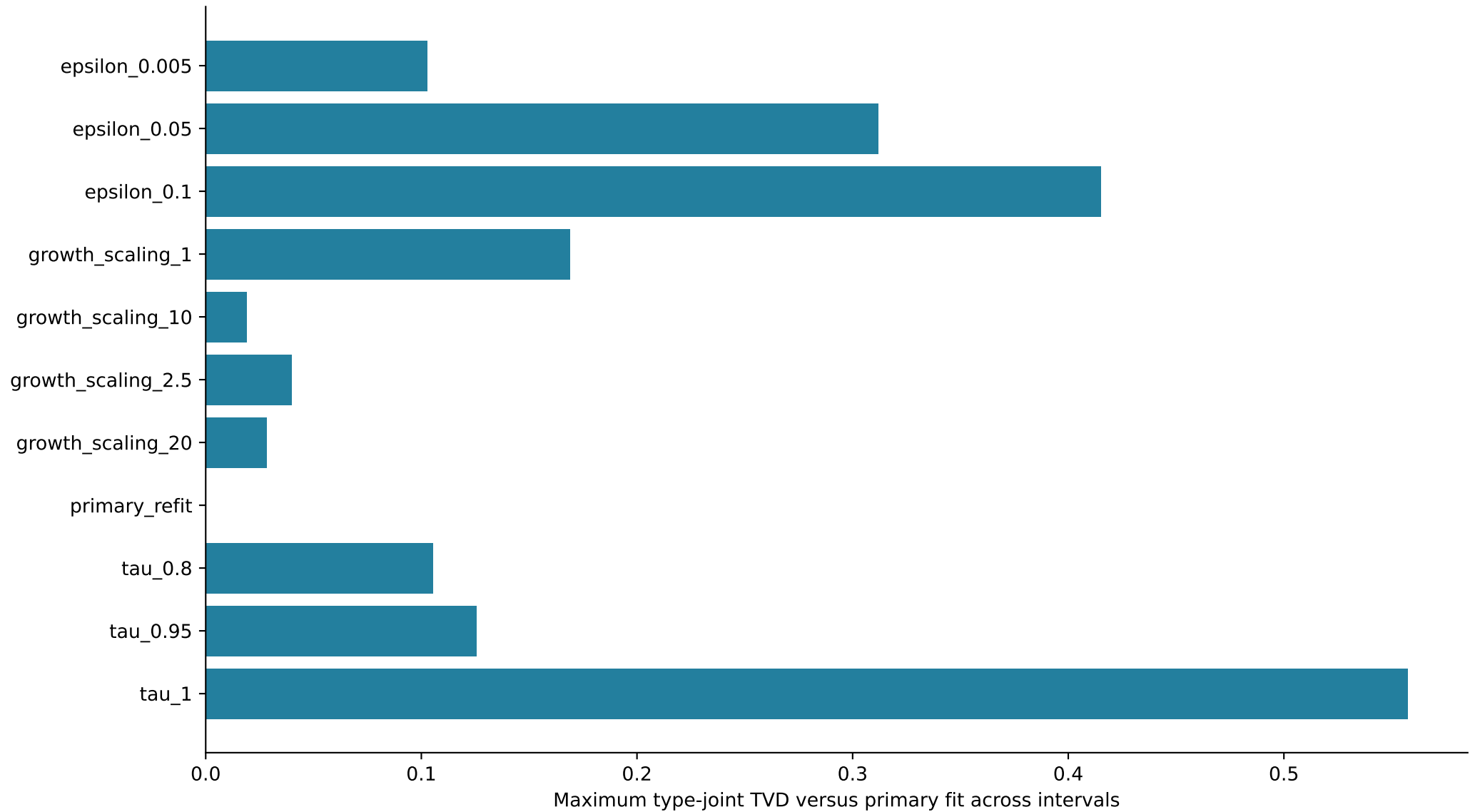
Top: aggregate source-mass effect. Bottom: mean within-fish normalized effect with descriptive 95% fish-bootstrap intervals; these are different estimands. No bootstrap refitting. The 0.05 line is an operational reporting rule, not biological significance.

3. Directional agreement is limited to two intervals



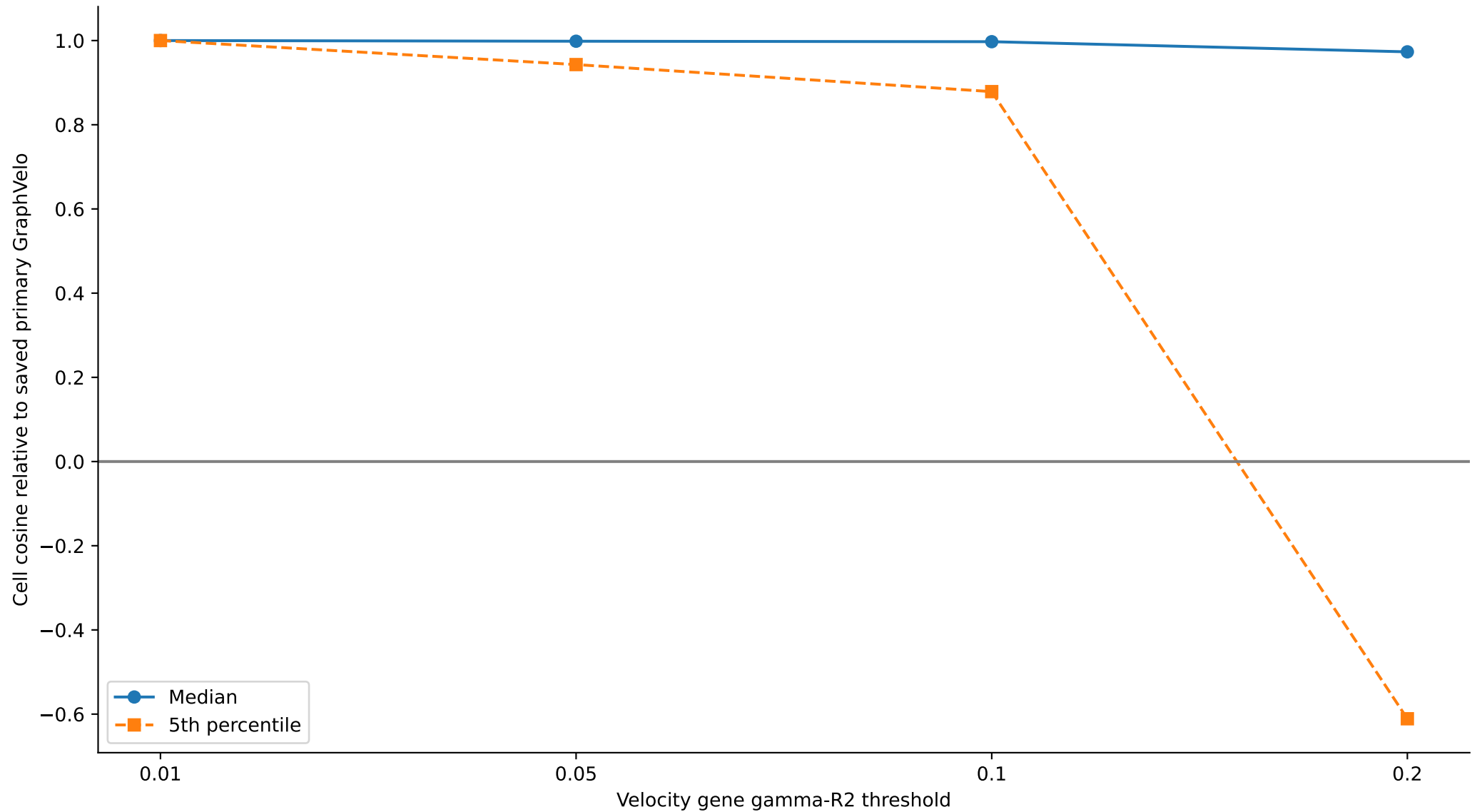
Green: positive absolute direction, positive fish-bootstrap lower bound, and within-fish permutation FDR < 0.05 (16-19 and 48-72 hpf).
Gray: criteria not met. Late negative cosine is not positive concordance even when better than the shuffled null. Shared expression/PCA means this is complementary computational evidence.

4. Solver choices materially affect the coupling



Plotted TVD is half the saved normalized type-joint L1 distance. Eleven one-factor configurations; all fits converged. Numerical convergence does not establish biological accuracy. Balanced tau=1 is a stress test. Source: native_moscot_parameter_sensitivity_summary.csv.

5. Stringent velocity-gene filtering destabilizes some cells



Thresholds 0.05 and 0.10 largely preserve primary directions. At 0.20, the lower tail becomes negative. Similarity to the primary fit measures stability, not correctness. Source: graphvelo_gene_qc_sensitivity_summary.csv.

6. What succeeded, what did not, and what is still unknown

WHAT THE CURRENT WORK SUPPORTS

- A reproducible saved native M0/M1 run, plus fish-aware audits.
- A larger growth-prior source-mass effect at 120–240 hpf.
- Positive complementary direction evidence at two intervals.
- Modest human expression-program concordance, with negative specificity tests.

WHAT REMAINS UNRESOLVED

- Actual division/death rates and true parent-descendant relationships.
- Negative absolute late-interval velocity agreement.
- Strong epsilon/tau sensitivity and sparse early sampling.
- Whether growth priors improve independent predictive performance.

HISTORY IS INCLUDED, NOT HIDDEN

- Earlier combined/reweighted models and smaller-marker-panel results.
- Cluster-23 exploratory trajectories, including mixed lineage markers.
- Partial notebooks and saved errors, explicitly labeled as historical.